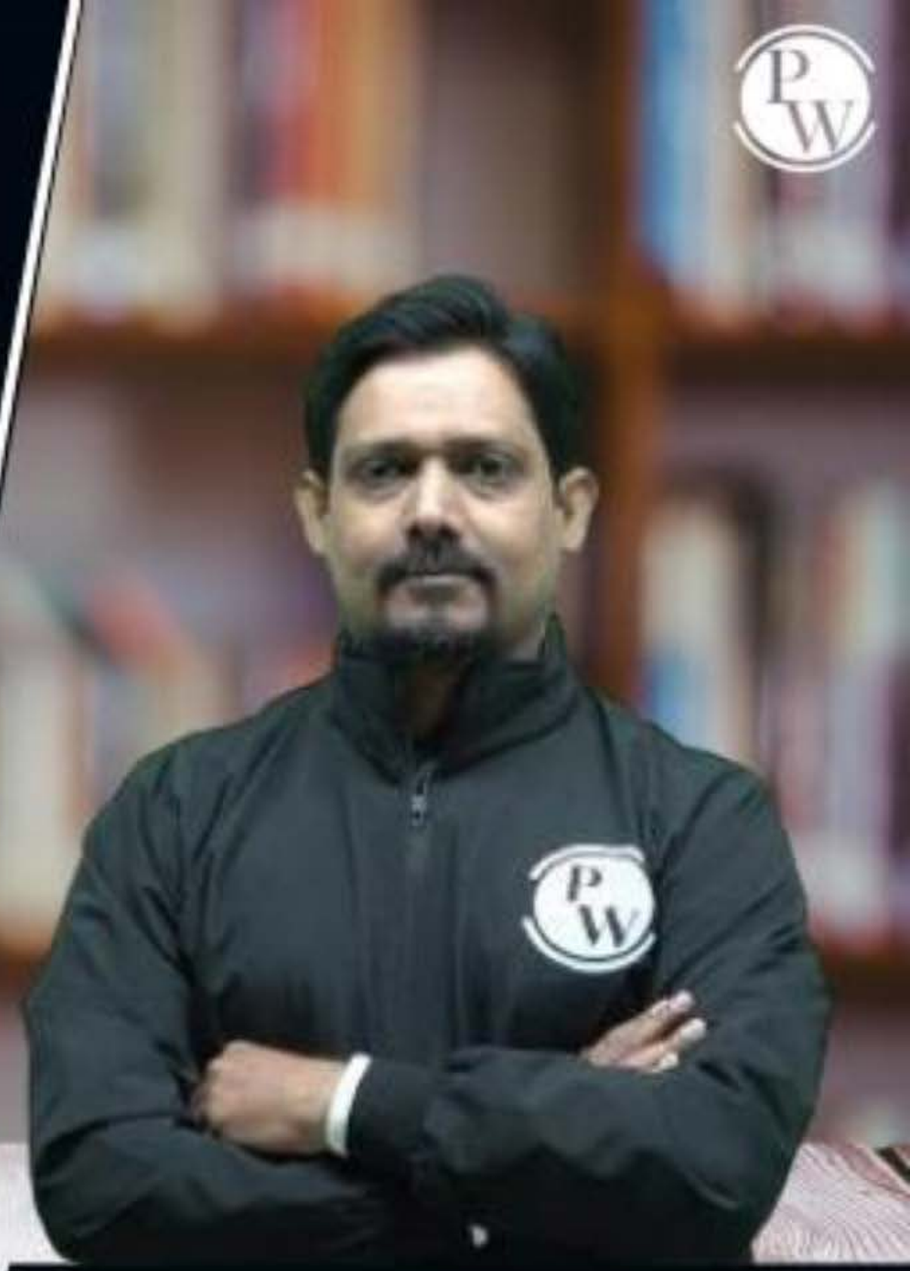
Computer Science & IT



Data Structure & Programming

Tree

Lecture No. 03



By- Abhishek Sir

Recap of Previous Lecture



Topic

Topic

Topic

Topic

Topic

Binary tree – FBT

Skewed tree

k-ary tree $L = I(k-1)+1$

k-ary Minimum : $kh+1$

Maximum : $\frac{k^{h+1}-1}{k-1}$

Slide

Topics to be Covered



- Topic — Counting of unlabeled & Labeled tree
- Topic — Binary tree Theorem
- Topic — Representation
- Topic — Traversal
- Topic

Slide



Topic : Tree

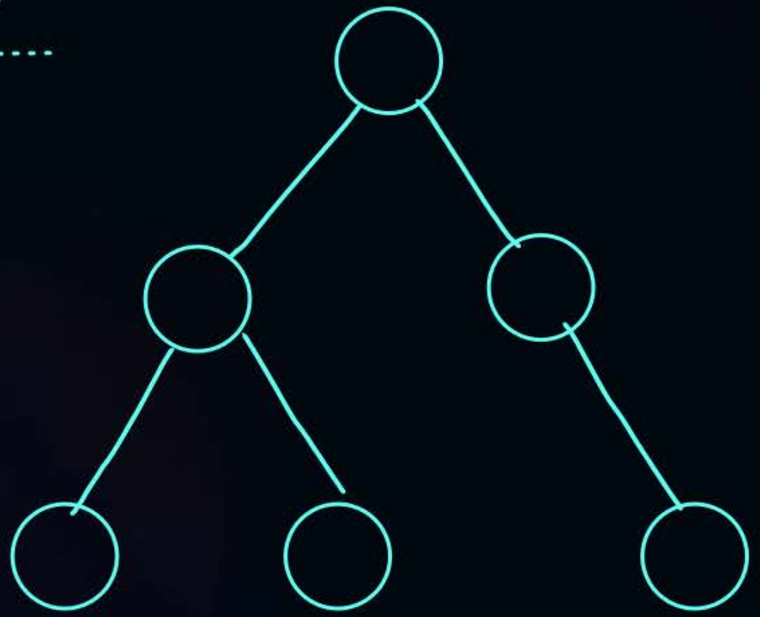
Theorem
Tree representation
Traversal

Slide



Topic : Tree

Unlabeled Tree



Slide



Topic : Tree

(1) n=1



(2) n=2



L	R
0	1
1	0



One Node is Root Node

Remaining Node is 1 Node

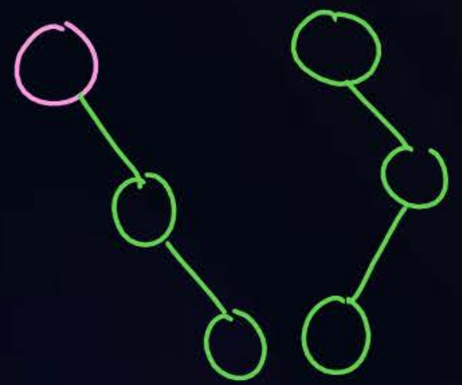
we can distribute in only 2 ways



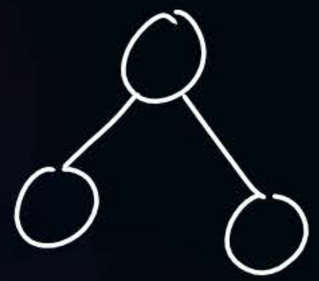
Topic : Tree

③ $n = 3$ ○ Remaining is 2 (one Node is Root)

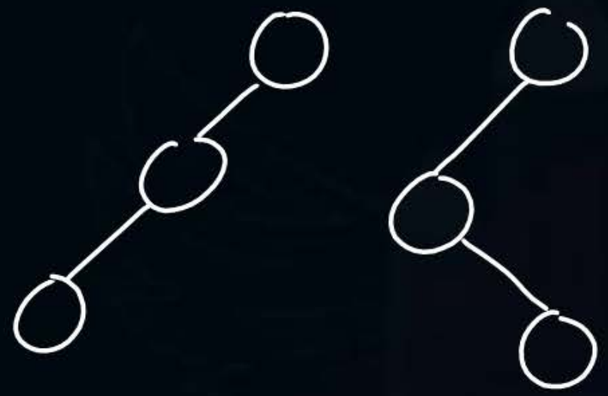
L	R
0	2



L	R
1	1



L	R
2	0



Slide



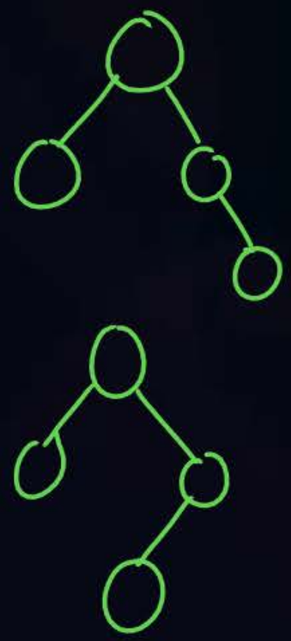
Topic : Tree

(4) n=4 0 (Remaining Node is 3)

L	R
0	3



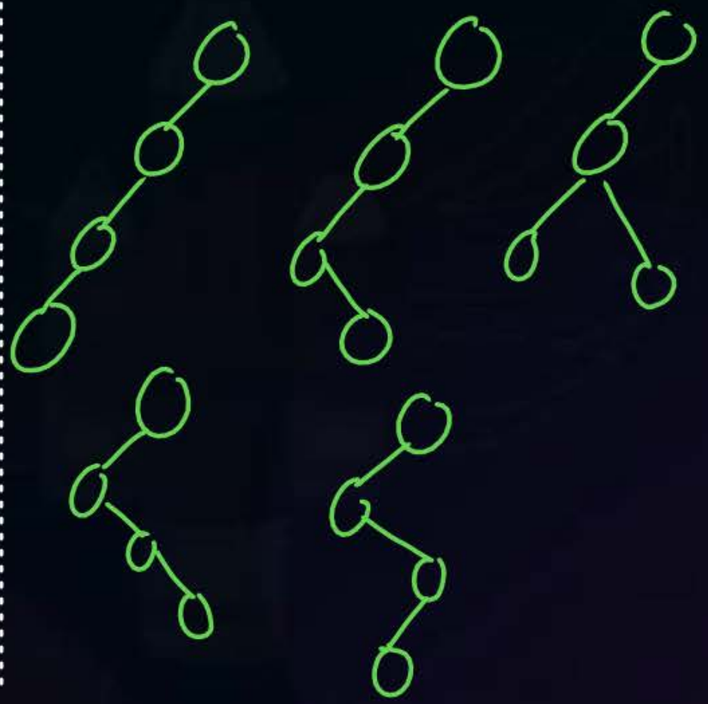
L	R
1	2



L	R
2	1



L	R
3	0



Slide



Topic : Tree

No. of unlabeled structure is Catalan No.

$$C_n = \frac{1}{n+1} {}^{2n}C_n$$

Slide

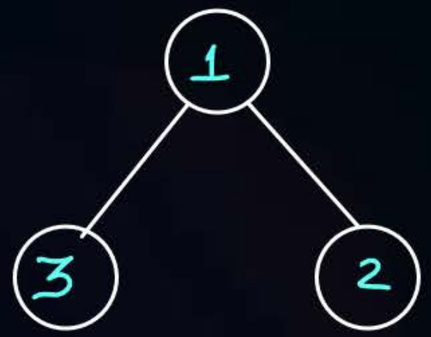


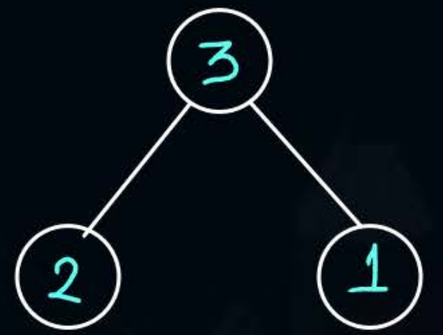
Topic : Tree

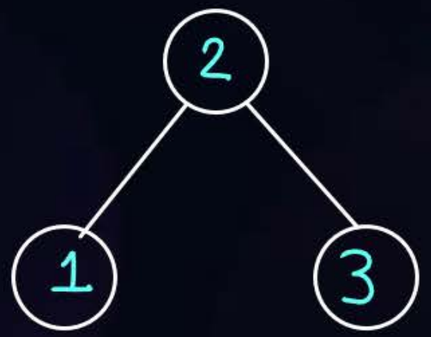


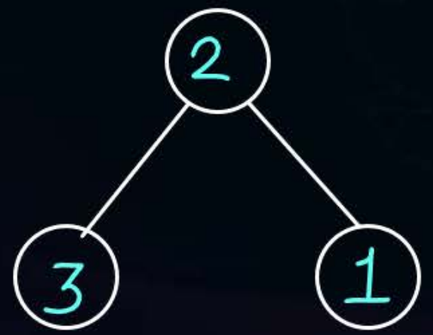
$3 \times 2 \times 1 = 3!$

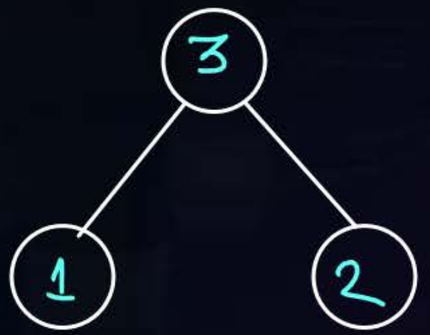
Label it with 1,2,3.











Slide



Topic : Tree

No. of Labeled structure is $\frac{1}{n+1} {}^{2n}C_n * n!$

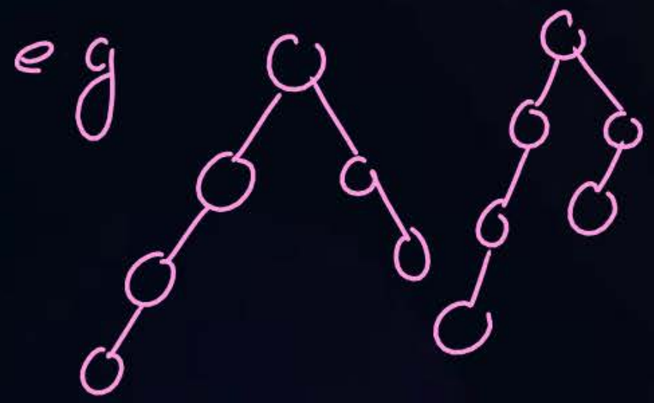
Slide

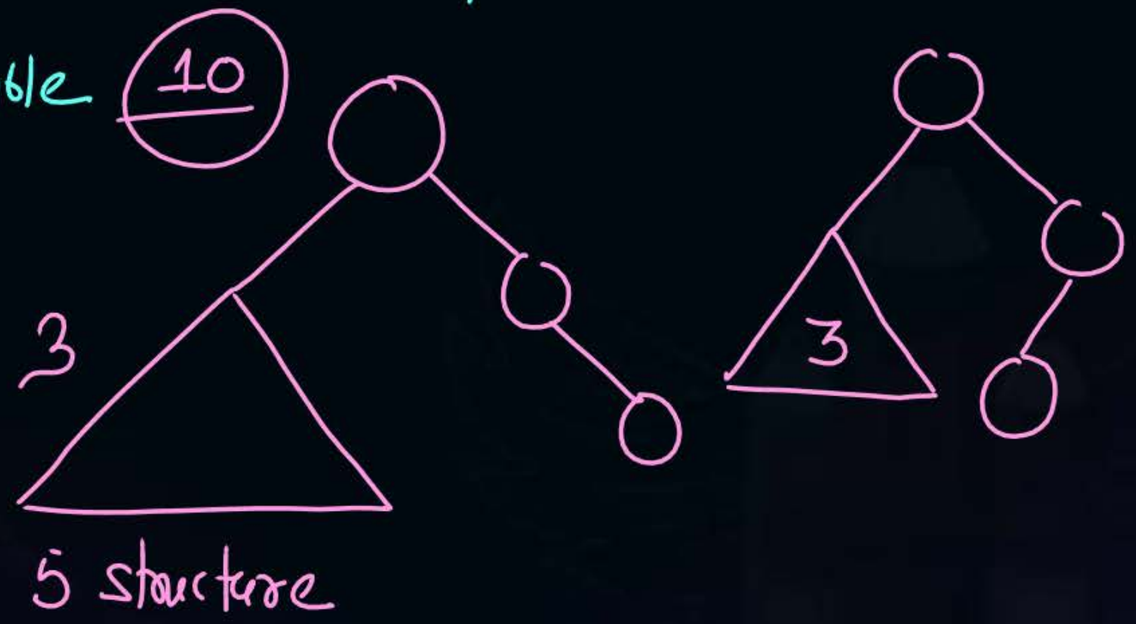


Topic : Tree

#Q Suppose binary tree is constructed with No 1,2,3,4,5,6 Such that Left subtree consists of 3 Nodes, How many unlabeled structures are possible (10)

eg





5 structure

Slide



Topic : Tree

#Q Suppose binary tree is constructed with No 1,2,3,4,5,6 Such that Left subtree consists of 3 Nodes, How many unlabeled structures are possible.

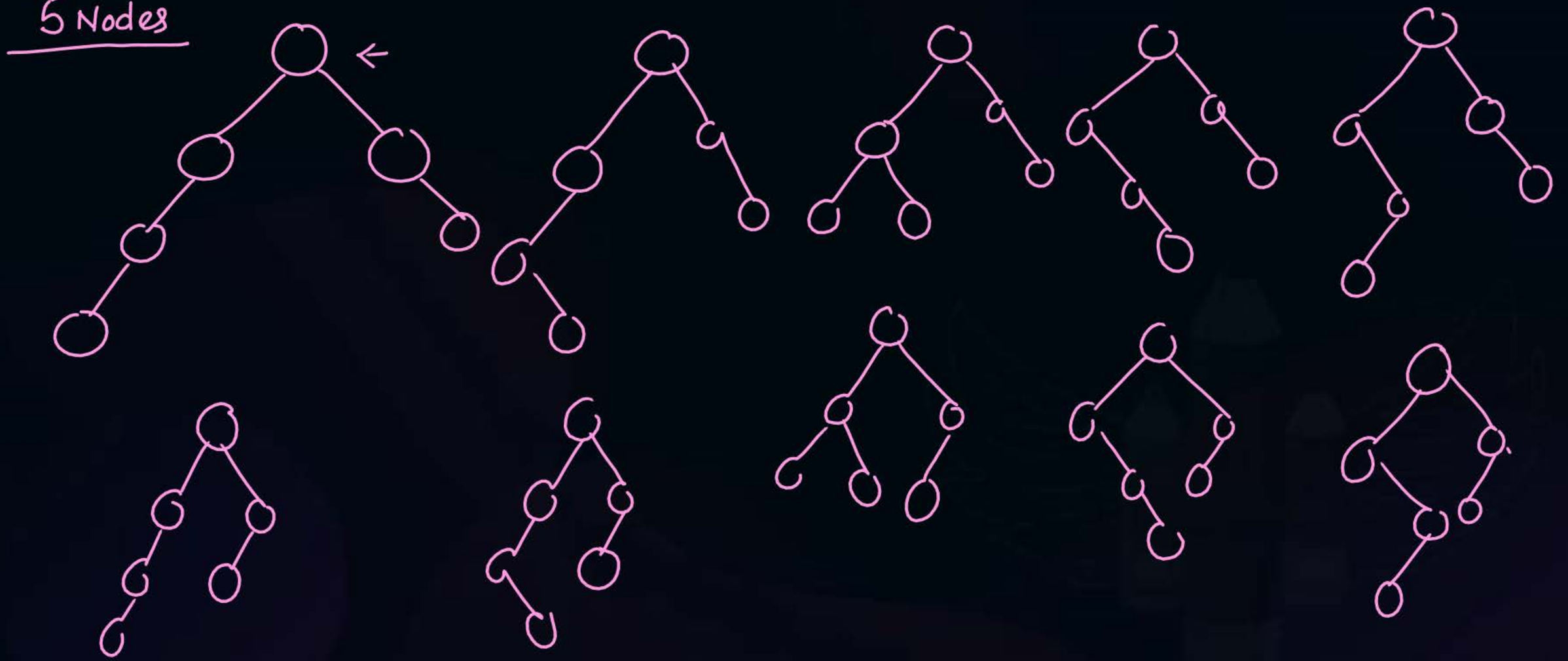
Slide



Topic : Tree

6 Nodes

5 Nodes



Slide



Topic : Tree

Binary tree Theorem : In a given binary tree if n_0 is No. of leaf Nodes and n_2 is No. of Node with 2 children then

$$n_0 = n_2 + 1$$

$n_0 : 1$

$n_2 : 0$



$n_0 = 4$

$n_2 = 3$



$n_0 : 2$

$n_2 = 1$



$n_0 = 2$

$n_2 = 1$



Slide



Topic : Tree

total No. of Nodes $n_0 + n_1 + n_2$

total No. of edges : $n_0 + n_1 + n_2 - 1$

n_0 - No. of Leaf Nodes

n_1 - No. of Node with 1 children

n_2 - No. of Node with 2 children

degree is No. of children a Node is having

$$n_0 \times 0 + n_1 \times 1 + n_2 \times 2 = n_0 + n_1 + n_2 - 1$$

$$n_0 = 2n_2 - n_2 + 1$$

$$\boxed{n_0 = n_2 + 1}$$

Slide



Topic : Tree

Tree Representation

1. General tree : Leftmost child & right sibling representation

2. Binary tree : Sequential Representation

   Tree Node representation

Slide



Topic : Tree

Sequential Representation

we use linear Data structure array to

to store binary tree.

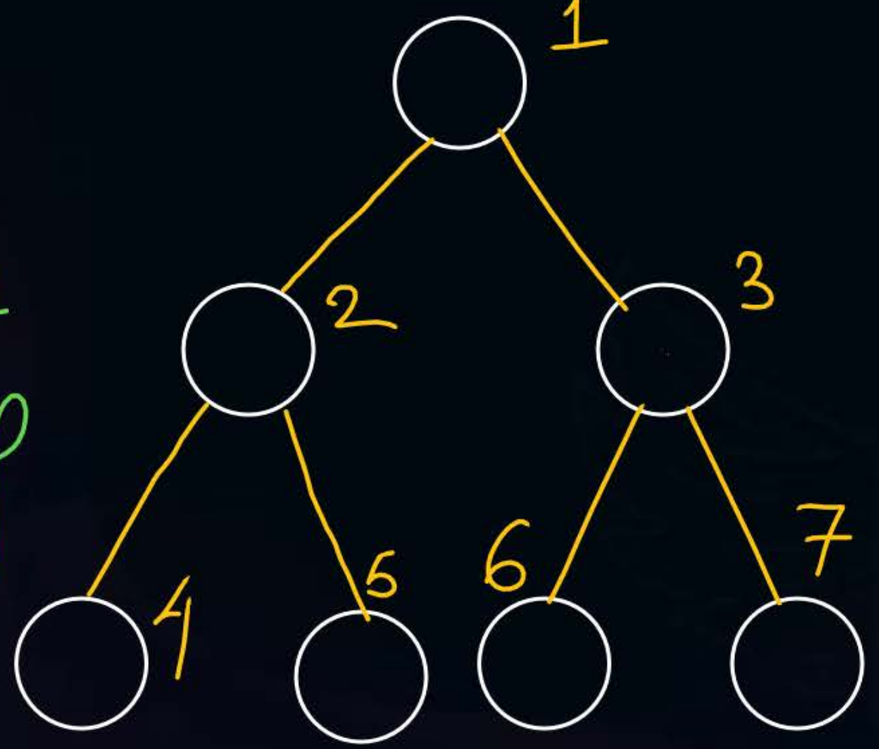
We assign Index

value to each Node

Such that the value of

Node stored in Index

position A[i]



Array is having

Index

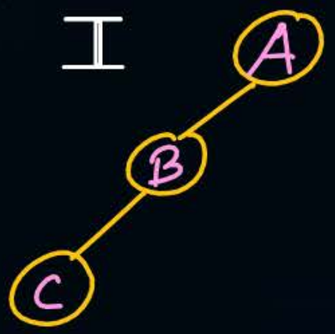
Application is in

Heap

Slide

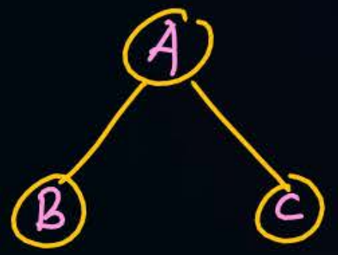
Topic : Tree

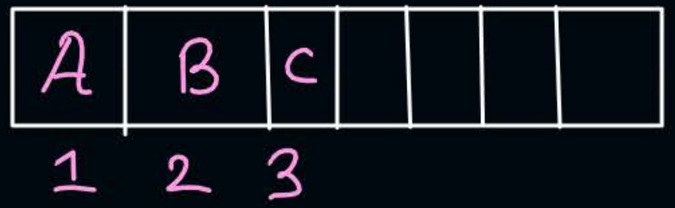
I





III





(V)



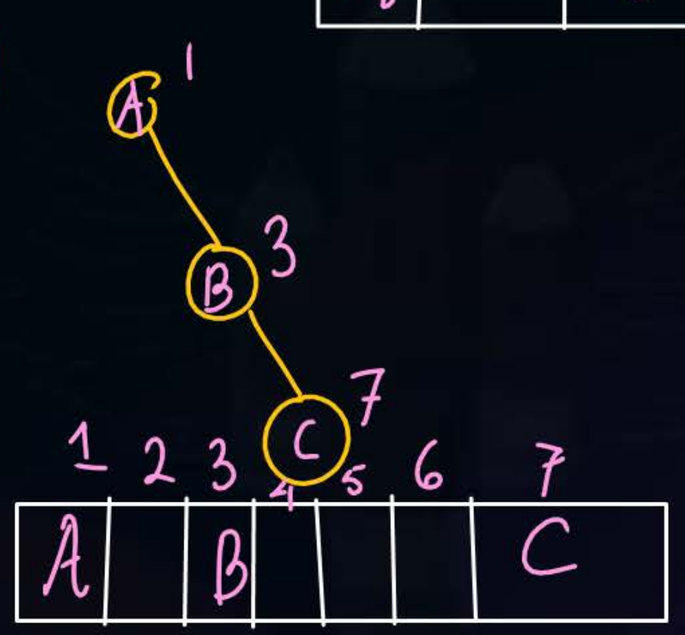


II





(IV)



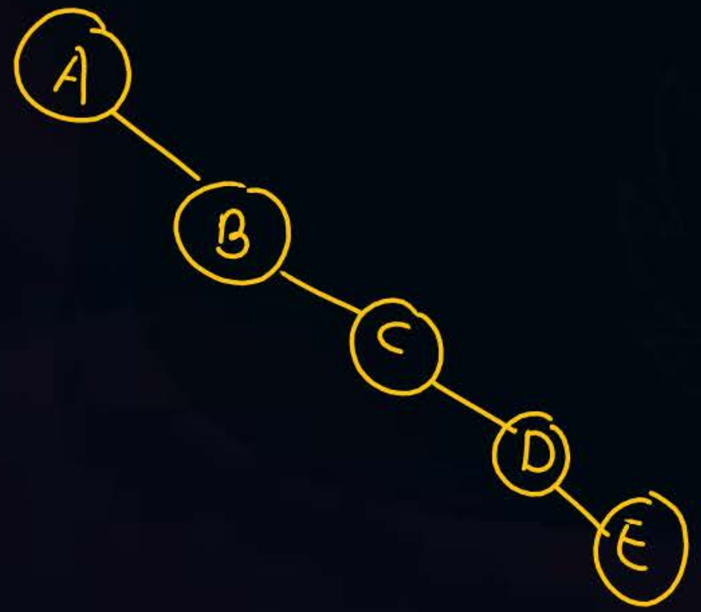
Slide



Topic : Tree

Minimum space for n node is n

#Q The Size of array required for Sequential representation of following tree



Array Size (31)

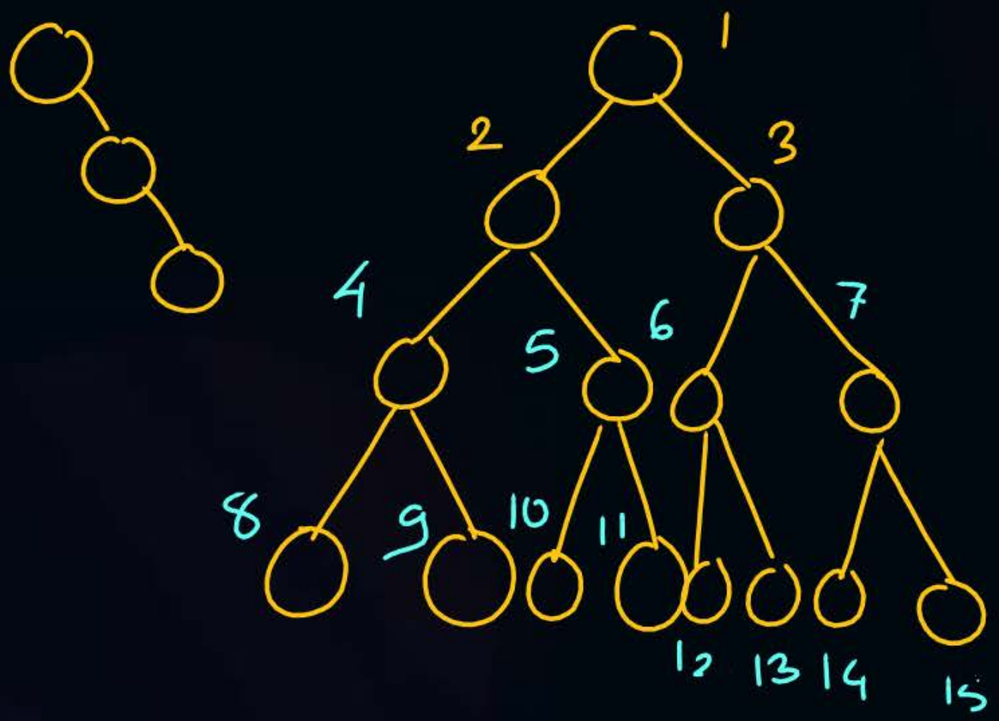
n · node

$2^n - 1$

Slide



Topic : Tree



3 Node – 7 $\Rightarrow 2^3 - 1$

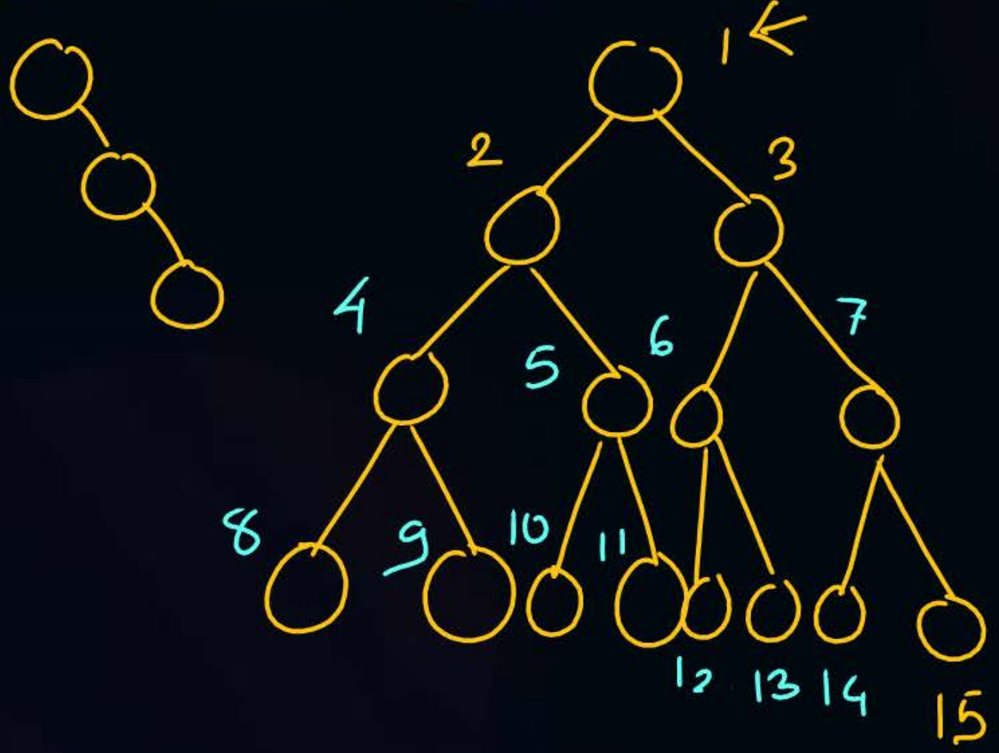
4 Node – 15 $\Rightarrow 2^4 - 1$

5 Node – $2^5 - 1 = 31$

Slide



Topic : Tree



If a Node is at index i ($i \neq 1$)

Index of left child : $2i$

Index of right child : $2i+1$

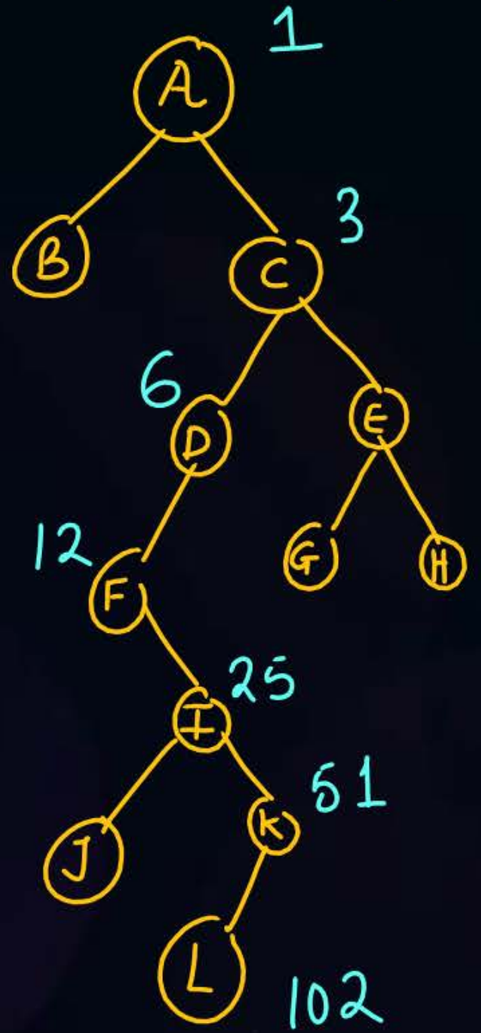
Index of parent : $\left\lfloor \frac{i}{2} \right\rfloor$

Slide



Topic : Tree

Q Consider following tree



if Sequential representation is used then index of Node L is

Index starts from 0

Index 1 start .. 102

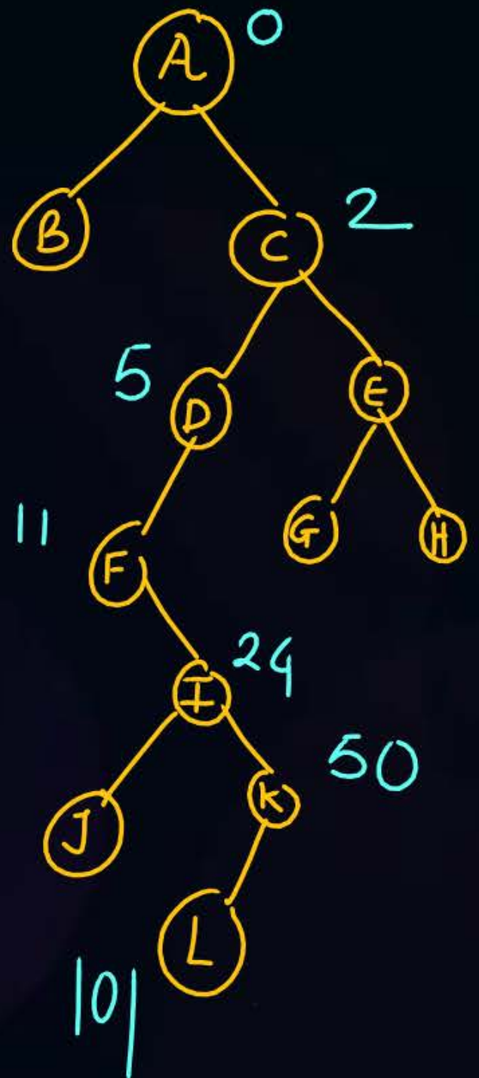
Index 0 start : 101

Slide



Topic : Tree

Q Consider following tree



if Sequential representation is used then index of Node L is

Index starts from 0

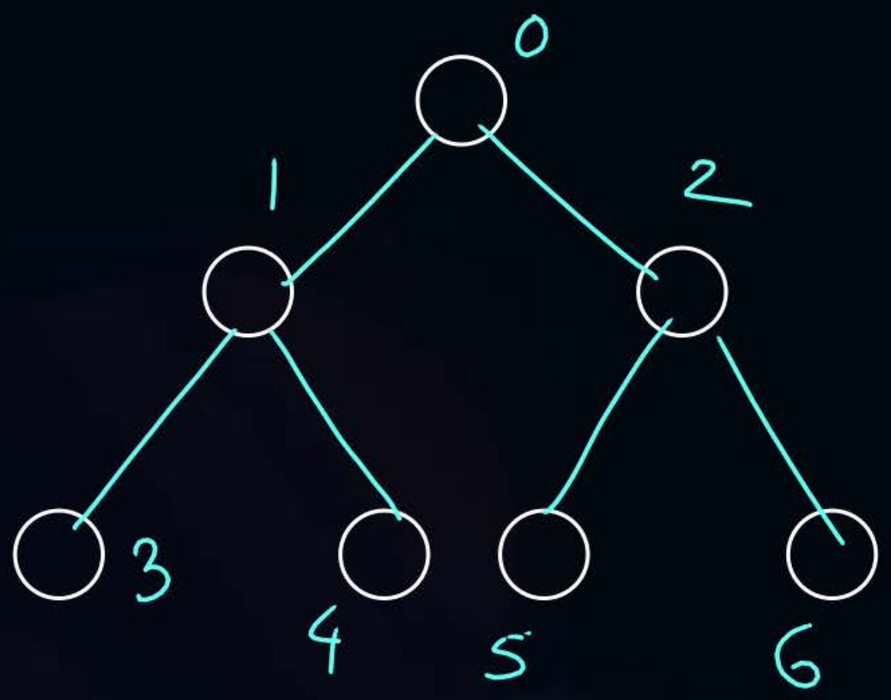
Index 1 start .. 102

Index 0 start : 101

Slide



Topic : Tree



if a Node at index i

Left child :- 2i+1

Right child · 2i+2

Slide



Topic : Tree

General Tree Node structure

Leftmost child & right Sibling representation

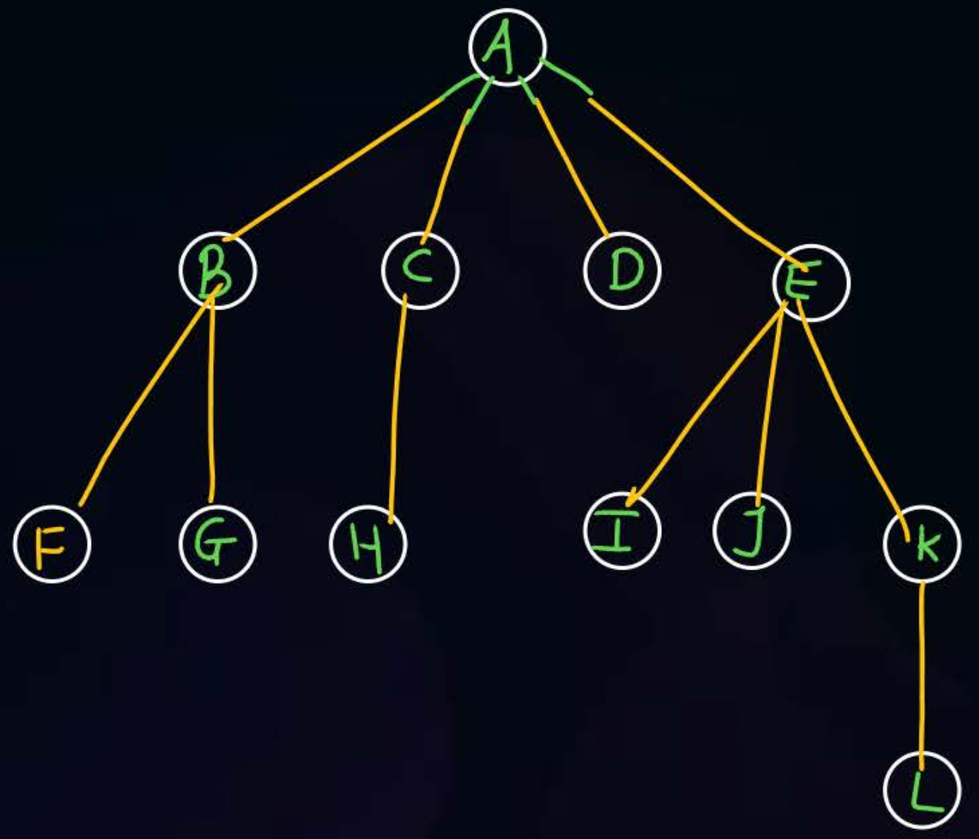
```
typedef struct GTnode{
    int data,
    struct GTnode *lmchild;
    struct GTnode *rsibling,
} GTnode,
```

Slide



Topic : Tree

General Tree Node Structure



A - leftmost :- B

A - right sibling :- NULL

B : leftmost child : F

B right sibling : C

G : leftmost child : NULL

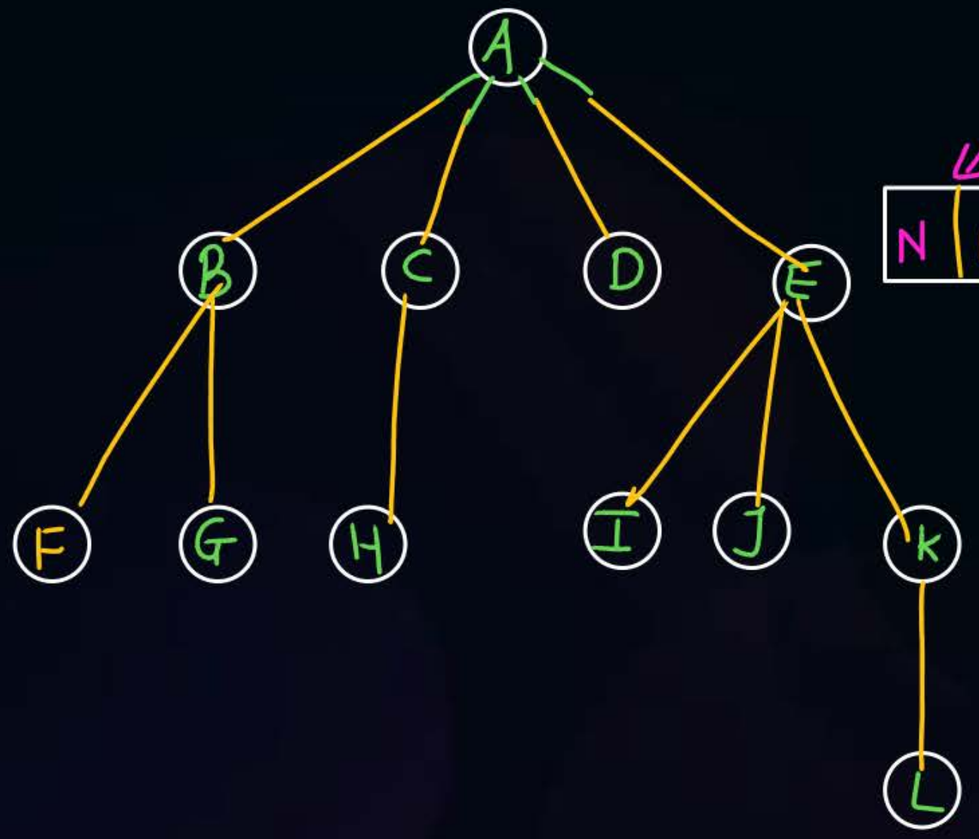
right sibling NULL

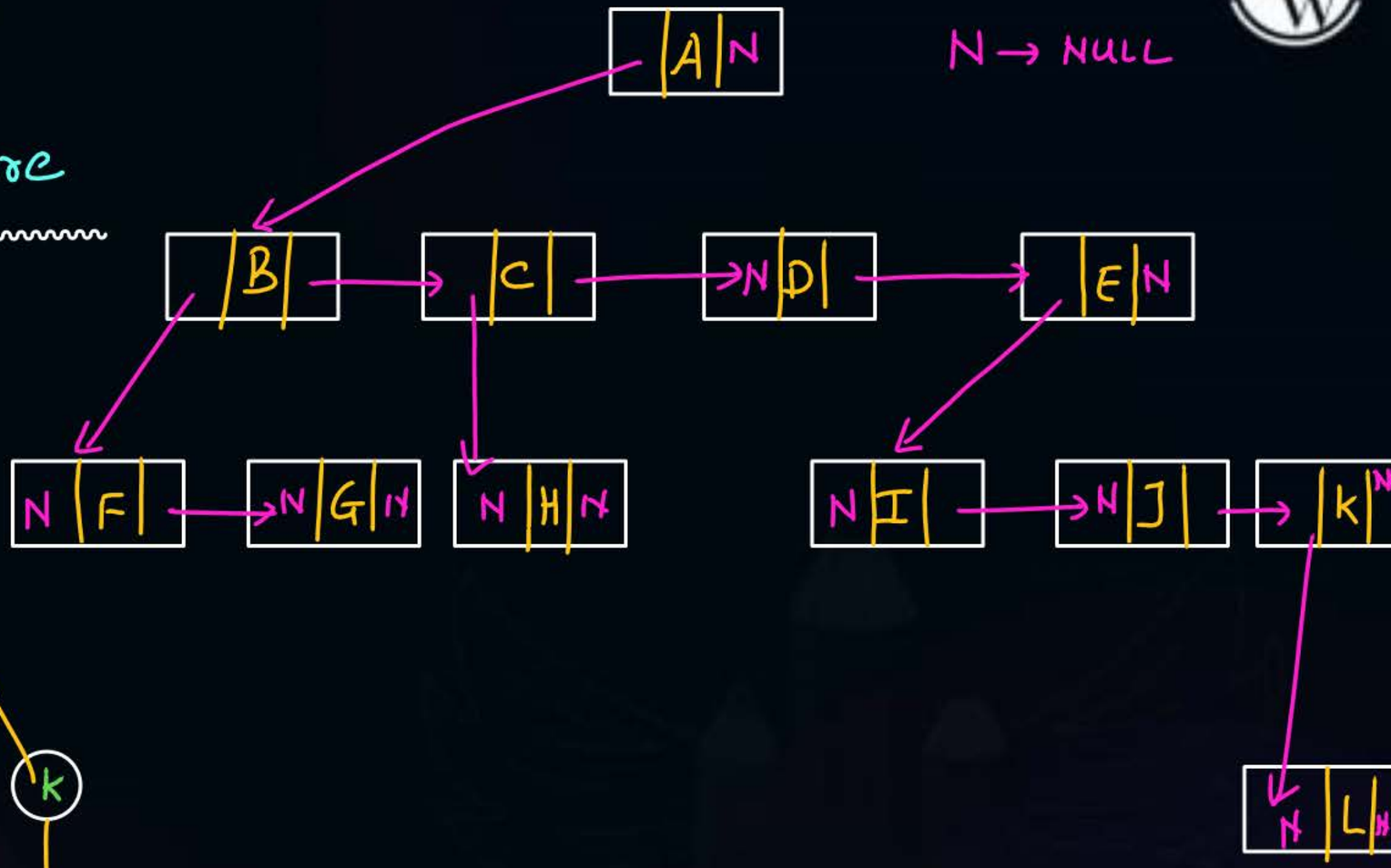
Slide



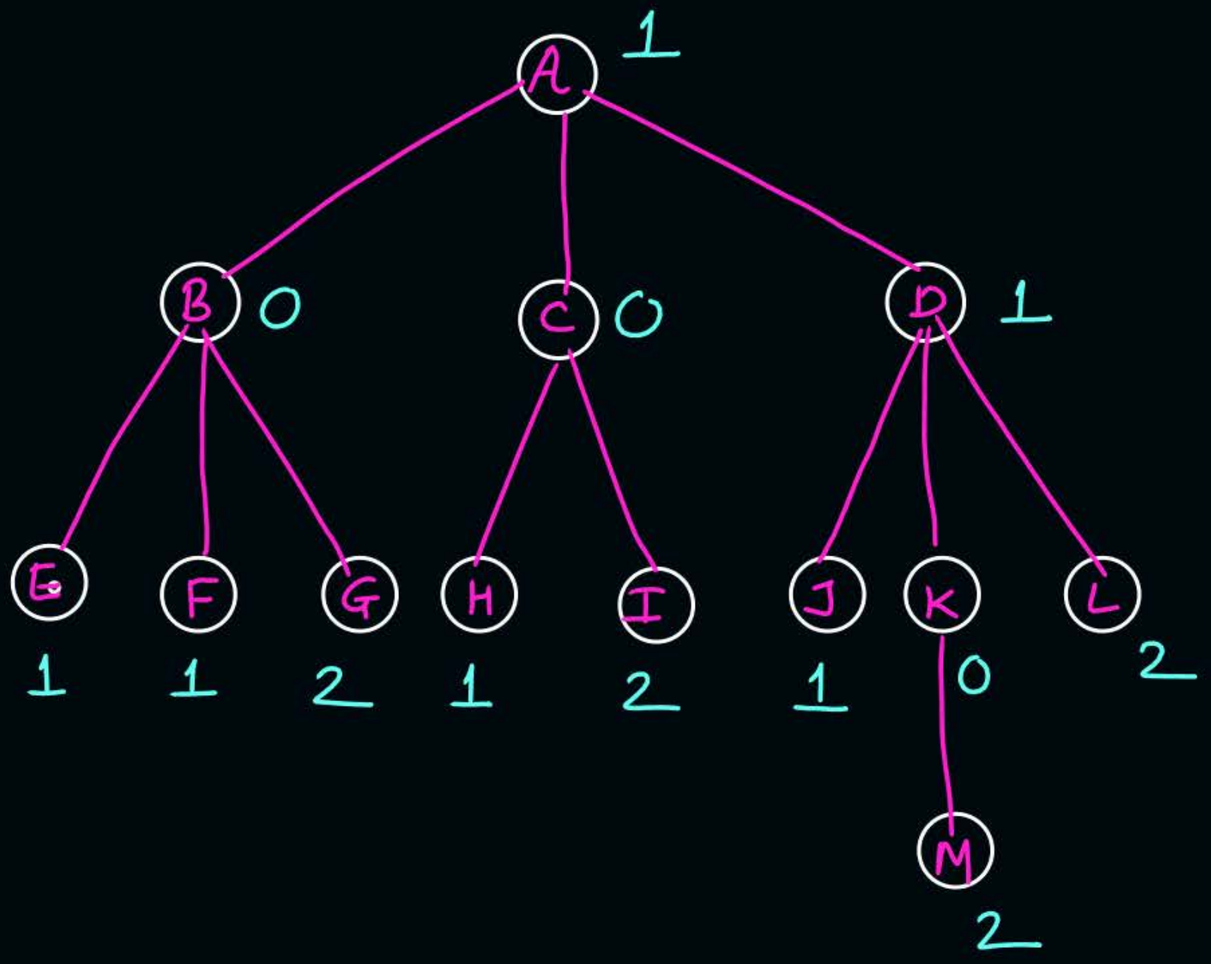
Topic : Tree

General Tree Node Structure





Slide



No. of NULL pointer present for Given general tree in Left most child & right sibling representation



2 mins Summary

Topic Counting of Labeled & unlobeled

Topic Binay tree representation

Topic General-tree Representation

Topic

Topic

THANK - YOU